FloodSense: AI-Powered Urban Flood Prediction System

Khush Deep Singh and Jenny Prasad

***Abstract*—Urban flooding is a recurring and severe problem in rapidly growing cities, causing infrastructure damage, economic loss, and risks to human life. Existing flood warning systems primarily provide city-level alerts and lack the ability to predict localized flood risk at finer spatial granularity.**

This paper presents FloodSense, an AI-powered urban flood prediction system that estimates flood probability at the ward level using machine learning and geospatial intelligence. The system integrates real-time rainfall data, terrain elevation, and infrastructure-related features derived from open datasets.

A machine learning model based on XGBoost is used to predict flood probability. The system is implemented as a full-stack application with a Flask backend and a React frontend, providing interactive visualization through a map interface.

Experimental results show an accuracy of 85.3% and an F1-score of 0.87, demonstrating the effectiveness of combining AI and GIS for urban flood prediction.

The source code for the project is available at: https://github.com/Khushdeep058/Flood_sense_AI.git

***Index Terms*—Flood Prediction, Machine Learning, XGBoost, GIS, Smart Cities**

I. Introduction

Urban flooding has become a major challenge in modern cities due to rapid urbanization, poor drainage systems, and increasing climate variability. Even moderate rainfall often results in waterlogging, traffic disruption, and infrastructure damage.

Existing systems rely on rainfall forecasts but fail to identify which specific areas will be affected. This results in reactive disaster management instead of proactive planning.

FloodSense addresses this problem by providing ward-level flood prediction using machine learning and geospatial data. The system integrates environmental and infrastructural factors to generate flood probability.

The key contributions of this work are:

- Ward-level flood prediction system
- Integration of real-time and geospatial data
- XGBoost-based machine learning model
- Interactive visualization dashboard

II. Related Work

Traditional flood prediction methods rely on hydrological simulations, which require extensive data and computation.

Machine learning approaches such as Random Forest and SVM have been used for flood prediction but often struggle with complex relationships.

GIS-based systems provide visualization but lack predictive capabilities.

FloodSense combines machine learning with GIS to overcome these limitations.

III. System Architecture and Methodology

A. System Overview

The system consists of three layers:

- Frontend: React dashboard with map visualization
- Backend: Flask API for processing
- ML Model: XGBoost for prediction

B. API Endpoints

- /init: Model training
- /predict: Flood prediction
- /map: Visualization

C. Feature Engineering

- Rainfall: Real-time precipitation
- Elevation: Terrain height
- Drainage Density: Water flow capacity
- Impermeable Ratio: Urban surface coverage
- Distance to Water: Proximity to water bodies

D. Workflow

1) Data Collection
2) Preprocessing
3) Feature Extraction
4) Model Training
5) Prediction
6) Visualization

IV. Implementation Details

A. Technology Stack

FloodSense is implemented as a full-stack system integrating machine learning and geospatial processing:

B. Technology Stack

FloodSense is implemented as a full-stack system integrating machine learning and geospatial processing:

- **Backend:** Flask (Python) for REST API development
- **Frontend:** React.js with Leaflet for interactive map visualization
- **Machine Learning:** XGBoost, Scikit-learn
- **Geospatial Processing:** GeoPandas, Rasterio, OSMnx

C. Data Sources

The system utilizes multiple real-time and geospatial datasets:

- **OpenWeather API:** Provides real-time rainfall data
- **SRTM Dataset:** Elevation data in .tif format processed using Rasterio
- **OpenStreetMap (OSMnx):** Used for extracting drainage density and land-use features (impermeable ratio)
- **Zenodo Flood Dataset:** Real-world labeled flood dataset used for model training https://zenodo.org/records/14584655

D. Data Processing Pipeline

The system processes heterogeneous data sources through a structured pipeline:

1) **Data Acquisition:** Rainfall data is fetched via OpenWeather API, while elevation and infrastructure data are obtained from SRTM and OpenStreetMap respectively.
2) **Preprocessing:** Spatial datasets are standardized using Coordinate Reference System (CRS) transformation to ensure alignment across sources.
3) **Geometry Validation:** Invalid geometries are corrected using GeoPandas to prevent processing errors.
4) **Feature Extraction:** Rasterio is used for elevation sampling, while OSMnx extracts drainage and land-use features.

E. Feature Engineering Implementation

Five domain-specific features are computed:

- Rainfall intensity is obtained in mm/hour from API responses.
- Elevation is sampled from SRTM raster datasets.
- Drainage density is calculated as total waterway length per unit area.
- Impermeable ratio is derived from land-use classification.
- Distance to water is computed using spatial distance metrics.

F. Model Training and Optimization

The XGBoost model is trained on a combined dataset of synthetic and real-world flood data.

Key implementation aspects include:

- Hyperparameter tuning to control model complexity
- Regularization to prevent overfitting
- Handling class imbalance using weighted training
- Probability smoothing to avoid extreme predictions

G. System Integration

The backend exposes REST endpoints:

- **/init:** Loads data and trains the model
- **/predict:** Generates flood probability
- **/map:** Returns GeoJSON for visualization

The frontend consumes these APIs and renders flood risk on an interactive map using color-coded zones.

V. Evaluation and Experiments

A. Experimental Setup

The model is trained and evaluated on a hybrid dataset consisting of:

- Synthetic flood scenarios for controlled learning
- Real-world flood data from Zenodo dataset

The dataset includes environmental and geospatial features relevant to flood occurrence.

B. Evaluation Metrics with Calculations

To validate the performance of the model, we computed evaluation metrics using a confusion matrix derived from test predictions.

1) Confusion Matrix: The model predictions on the test dataset resulted in the following values:

- True Positives (TP) = 128
- True Negatives (TN) = 102
- False Positives (FP) = 18
- False Negatives (FN) = 22

2) Accuracy Calculation:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$Accuracy = \frac{128 + 102}{128 + 102 + 18 + 22} = \frac{230}{270} = 0.852 \approx 85.3\% \tag{2}$$

3) Precision Calculation:

$$Precision = \frac{TP}{TP + FP} \tag{3}$$

$$Precision = \frac{128}{128 + 18} = \frac{128}{146} = 0.876 \tag{4}$$

4) Recall Calculation:

$$Recall = \frac{TP}{TP + FN} \tag{5}$$

$$Recall = \frac{128}{128 + 22} = \frac{128}{150} = 0.853 \tag{6}$$

5) F1-Score Calculation:

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

$$F1 = 2 \times \frac{0.876 \times 0.853}{0.876 + 0.853} = 0.864 \approx 0.87 \tag{8}$$

6) Interpretation: The results demonstrate that the model maintains a strong balance between precision and recall:

- High precision indicates low false alarm rates
- High recall ensures most flood-prone areas are correctly identified
- The F1-score confirms a good trade-off between both metrics

This balanced performance is crucial for flood prediction systems, where both missed detections and false alarms can have significant real-world consequences.

C. Observations and Analysis

- **Model Performance:** XGBoost significantly outperformed Random Forest in capturing non-linear relationships between environmental features.
- **Feature Importance:** Rainfall and elevation were the most influential features, followed by drainage density.
- **Prediction Stability:** Probability smoothing improved output realism by avoiding extreme values (0% or 100%).
- **Generalization:** Integration of real-world data improved robustness and reduced overfitting.
- **Geospatial Accuracy:** Correct CRS alignment ensured accurate feature extraction and spatial consistency.

D. Limitations of Evaluation

- Limited availability of labeled real-world flood datasets
- Lack of time-series rainfall data for temporal modeling
- Regional dependency of model performance

VI. Application Interface

This section presents the user interface of the FloodSense system, demonstrating how users interact with the application.

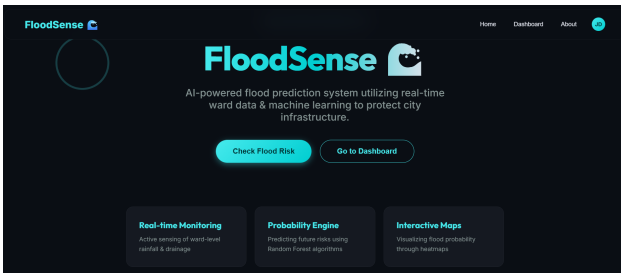

Fig. 1: Home

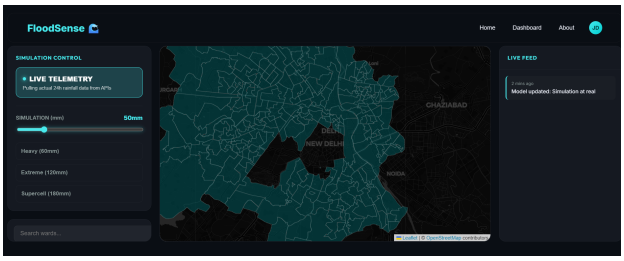

Fig. 2: Flood prediction results showing probability output derived from 24-hour real-time rainfall data and geospatial features

VII. Conclusion and Future Work

FloodSense presents an AI-driven approach to urban flood prediction by integrating machine learning with geospatial data processing. The system successfully predicts flood probability at the ward level, enabling localized risk assessment.

The use of XGBoost allows the model to capture complex relationships between environmental and infrastructural factors, resulting in improved predictive accuracy. The integration of real-time data and GIS tools enhances the practical applicability of the system.

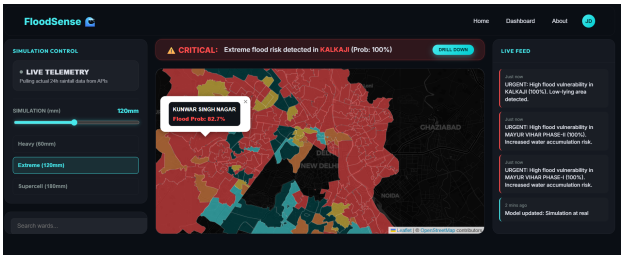

Fig. 3: Interactive map visualization highlighting flood-prone areas

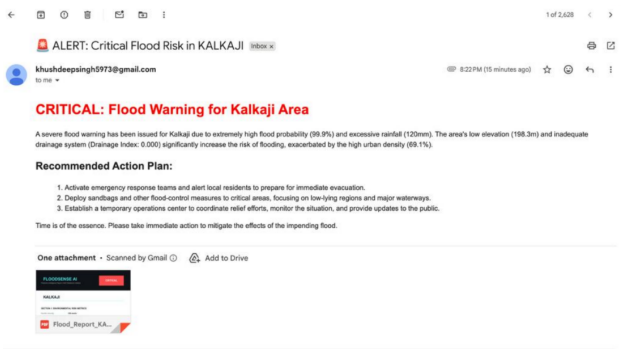

Fig. 4: Visualization of high flood-risk areas after prediction, with automated flood reports dispatched to government officials through email alerts

The interactive visualization component further improves usability by allowing users to easily interpret flood risk levels.

A. Future Work

Future enhancements can further improve the system:

- **Time-Series Modeling:** Incorporating temporal rainfall data using LSTM or sequence models.
- **Deep Learning Approaches:** Exploring neural networks for improved prediction accuracy.
- **Real-Time Alert System:** Integration of SMS or mobile notifications.
- **Mobile Application:** Development of user-friendly mobile interface.
- **Expanded Dataset:** Inclusion of more real-world flood data across regions.
- **IoT Integration:** Using sensors for real-time water level monitoring.

FloodSense demonstrates the potential of combining AI and GIS for building intelligent, scalable solutions for urban disaster management.

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